ActInf Textbook Group ~ Cohort 6 ~ Session 6 (Chapter 4, Part 2) ~ 3/25/2024

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all right greetings thank you everyone for joining we're in our second discussion on chapter four let's just jump to it um anyone can write a question in the chat or raise their hand or just like unmute and go for it and it could be like anything you remember from the chapter or like a part you liked or we're curious about or just any any question that you want to jump in with like any any section that people prefer just un unmute just any page any section oh Aon go for it so all um I I was really scratching my head trying to compare figure 44 with figure 46 um there's like a very large amount of overlap between the two sets of diagrams but there's like some very uh so I feel like there's like a clear parallel setup um there's some edges that are different between the two and I was like trying to understand like what do those changes mean um and like right these were yeah uh and in both cases I think especially the left hand side diagram is like virtually identical um between 44 and 46 I think I have the numbering that right um but uh yeah I I don't know if that's like too much to fit into a question but I was I I thought the difference there must be meaningful but I I didn't fully I couldn't make sense of it it's a great question like for sure on first pass it's like two cities like maybe there's some differences and which roads and what's connected to what but at a first pass like they are both very very similar um so a few things are being shown here um to to kind of give context on this qu question let's pull back to figure 43 so here we see this kind of like Rosetta Stone figure with a basian graph for the discrete time model and a basian graph for the continuous time model on the bottom they're kind of being like laid out like this to emphasize their structural similarity even though they're treating time differently which we can return to but the top is treating variables in terms of discrete sequences of events whereas The Continuous time is treating it in terms of higher and higher derivatives of a given um variable X the important thing though that makes these similar other than their structure is that in a base graph you have edges are variables and then the uh I'm sorry nodes are variables and the edges are causal connections okay so now we get to aasian message passing scheme so one of the really cool things about AAS graph is because the edges represent the sparsity of causal Connections in the model you only need to consider edges that are local to a node so it's like the traffic at a node is strictly and entirely defined by the edges leading in um so when constructing a given basian model like you specify the kind of causal architecture of the world and then messages are passed between the variables so this is going to be a slightly different base graph than the figure 4.3 but it's going to emphasize how certain kinds of architectures pass certain messages to each other and then how that enables different kinds of computations to happen like there's a different variable here so like right off the bat even though we see some variables that were in figure 4.3 like pi and S there's also different variables like the um Epsilon and the the ITA um okay so two things are being highlighted in this figure 4.4 on the left is going to be about hierarchy and nesting on the right is going to be about unfolding Dynamics through time so the motif on the left that describes the nesting is going to be like each unit of four is kind of like a level in a nested hierarchical model and basically what's happening is the E is calculating an error like Epsilon for error and then those errors are being passed up to a higher order in the um graphs so that is like the kind of classical predictive processing architecture we have strictly stacked levels and then what's happening within each level is it's doing a differencing and then passing the errors up that's the hierarchical motif The dynamical Motif is going through time so this one actually has more similarity to the discrete time in 4.3 and here it's kind of like a predictive processing but emphasizing the dynamic here's the sequence of observations coming in current time step the future time step past time step and then each of those observations are kind of meeting in the middle and calculating an Epsilon based upon the Collision of the incoming observation and then the hidden State prediction so like the Epsilon would be zero if the hidden State prediction was exactly what the observation was and then those are getting passed on up I'm um okay then 4.6 here the only difference is that the again the very similar shapes with like the motif of four and the passing up but now we're in the continuous time setting so that's why there's um more variables like we see in the continuous time World um and now rather than representing the past present and future time step just like the top here past present and future time step in 4.6 now they are representing different orders of the generalized ordinates of motion so the first the second and the third derivatives so 4.4 and 4.6 are kind of like transposed from figure 4.3 the top and the bottom but now it's representing each of those halves in terms of the message passing that it reflects computationally so what's cool about it is it shows again a deeper similarity between these two kinds of models what's kind of challenging about it is it's introducing new notation and it's not exactly the same model as before any thoughts on that or more more on that question or it could be a different question that anyone has we can go to any any part of four or like any any can just anyone can just unmute and go for it or write in the chat yes TCC it's kind of like Parallax in Vision like chapter 4 is presenting the generative models of active inference so in terms of where we are in the textbook like chapter one introduced it chapter one two and three approached active inference from the low road and high road and then chapter four is like actually the kernel of active inference and then the discreet and the continuous time are going to be unpacked more in chapter seven for the discreet and chapter eight for the continuous time however here they're being really like tightly jux opposed and interwoven again that speaks to a really deep point which is that time can be handled either in discreet or continuous fashion in active inference that's cool deep Point what's challenging is that even when they're discussing similar kind of cognitive functionalities the notation is very different between the two forms it would be confusing in a different way if the exact same notation was used because it wouldn't be clear whether we were talking about the discreete or the continuous time however this is kind of like a second set of vocabulary but we couldn't use exactly the same because some of the variables are slightly different but like using Y for observations in the continuous time and then using uh o for observations in discreet time that's just how they do it in this chapter so it's kind of like they're introducing both time treatments and emphasizing again and again and again that that both of these are like two flavors and then they are going to revisit it in the dedicated chapter seven and chapter 8 but it's kind of like a double barreled chapter because they're introducing generative model not in terms of like well here's a simpler generative model now we're going to go to more complex they're introducing like a dual stream like the two simplest forms but anyone can just unmute and go for it or we can look at the chapter 4 questions or whatever people prefer okay if no one else has I'll go again I I'm confused also about the the structure of um uh policy variables like okay so like a in a hidden State an observed State whatever like I can like sort of acknowledge those are sort of like things that exist in like some not fully specified space um policy especially because I think it's like trying to call back to reinforcement learning literature like I feel like in some what's what's like the type of that like if you were to describe like the space in which a policy either the discret version the pi or the uh continuous version The V like what that lives in uh or if you were to like implement it in a program like what what the type associated with that is like what's it shape great question Pi is a list of all policies so it's always a vector and it always sums up to one and that's really the critical move that allows us to talk about the likelihood of a policy that's why habit is the prior on policy and then expected free energy updates our prior on policy into a posterior because the shape of that variable is a list that sums to one so it's a probability distribution so affordances are the actions that can be taken in a moment so if we're like in a grid world the affordances will be like up down left right stay so it'll be like five then if we're only a single time Horizon a single time step agents affordances are the same as policy space because we're only selecting the next action so say we have five options if we had an even prior across those five it'd be like 0. 2 point2 point2 to and then that could be sampled from or updated into a policy posterior if we're talking about an agent that does planning then the shape of Pi is affordances raised to the power of the time steps so if we have five affordances and then we're doing two time steps then Pi is 25 long it's up up up down up right and so it's all of the two-way combinations so that's the kind of exponential explosion with policy planning which we can come back to but at a first pass the important things about Pi are that it's a list it sums to one which is what makes it a probability distribution and it's a distribution over policies that the agent is selecting from just just to clarify or check my understanding so um then whenever we see like uh V with the Tilda representing a trajectory over policies like that is maybe should be regarded as something that's like because as exponential and like a unbounded number of steps is like not something that we should consider like concretely representable in a program interesting so the The Continuous time setting doesn't do explicit planning it does so the the PDP at the top here this is where we have the kind of classical computer science exponential tree search like more and more discrete combinations you know if you had five times steps deep you'd have five to the fifth discrete options it's kind of like a chess algorithm um in the continuous time setting planning is not explicit like let's just say that we had some wild function that we were trying to approximate when doing this kind of a tailor series approximation like evaluate the the function at zero and then take the first derivative so let's just say we stopped there we're just going to evaluate the function and then look at the local slope that already expands to everywhere that doesn't mean it's going to be accurate everywhere but basically the tailor series does expand no matter how many levels deep you go it already implicitly includes basically infinite future planning but for explicit planning in a tree like setting that's where the discrete time comes into play in um in this page um Thomas P's discussion so early implementations were in The Continuous State space because this has the most similarity with like basian filtering approaches then people started wondering well how do we get sequential Dynamics then they moved into this kind of coupled dynamical systems space but then that was only getting so far because there wasn't an explicit representation of planning then they moved into the discrete planning environments those branches kind of came back together by nesting The Continuous Dynamics at the lower level with the discrete Dynamics at a higher level so you could do a lot of work in the continuous bace and then also have that like wrapped by a discrete planning step that's like a little bit about how the timeline worked but just to summarize in the discrete setting this is the most like classical planning in the continuous setting you don't have planning as such it's kind of like you have a car going around a turn in the discret um Step uh discret time world it's trying to estimate all the branching paths it could take in the continuous time world the driver's like I'm turning 20 degrees to the left that's like the first derivative of my turning angle and then I'm just I'm holding to that and then that curve continues so it's not explicitly saying at seven times St now here's where I expect to be so it's the strength and the weakness of the discrete time the weakness is that there's a exponential explosion of the tree just like a chess algorithm but the strength is that you can actually talk about the difference between up up up up up down and up up up up up right which you couldn't really get except by going super deep in the space so they both have these kind of like strengths and weaknesses and then that is resolved or addressed by building hierarchical nested models that have components that have continuous treatment of time and components with discreete good call um let's see in page 73 yes someone else has already added this but yes good call there's several typos in the book we can go to any of the written questions or like just any any remark about how anything that we've gotten up to or just like anything anything active related we can totally go for it because chapter four is just it's kind of like the heart of active inference um albeit described in a very um notational way David or anyone go for it uh yeah I was just wondering uh if there's any relationship to the partial gamma function in regards to how the branching um seems to work and fit backwards like if you're mapping back up through the stack of the decision tree that Maps back to the initial I just didn't see any of the the math there in the chapter but I've seen some correlation in some of the um the math that I've been working on recently um so if you go further down uh there's a section on branching and the relation yeah so Behavior near Branch point and the relation between the branches so I was just wondering if um if that fits in or if my intuition on the the correlation is incorrect there it's an interesting connection I don't know about like this analytic family but a few thoughts are inactive inference we're treating every moments as a branch point so that is the control decision slash question is the branch point is defined by the affordances that can be taken at that point and so there's some um you know there's some kayaking down the river where no matter which way you Branch it's like you're going to get kind of brought back in so like the long-term consequences don't matter but then there's other moments where to to go a little bit one way or the other is going to diverge so that is kind of like you have the agents on the kayak and they have the behavioral Branch point but of course just knowing that they can tilt left or right doesn't tell you about like the river which is kind of like the actual process unfolding so I'm not sure exactly how that would connect but action is a branch point another another possible connection there is like in this chapter we're seeing like the basic kernel like the minimal essential model that describes um either you know in either of these time treatments but then there's all kinds of other cognitive functionalities like what's known a sophisticated Act of inference is where in the moment you're not just planning um or or trying to estimate like what will I be doing and what will I see but you can also project out what will I be thinking then about the past so again that explodes the state space even further but it allows for like sort of prospective retrospective but that that kind of phenomena is not here but it's like a modification on this base model yeah I guess I was seeing it more in terms of um the primes and the double primes on the uh on the bottom set and maybe that was just my miscorrelation of just misreading the um terminology as it's as it's represented for the uh the lower States so you see I um y y Prime and then y double Prime there that's the first derivative and the second der right but in order to move back like if you were if you were collapsing back into the initial kernel stage wouldn't it have to still step through th those premises in that order or would it all collapse at the same exact time again there's a lot to this question I mean what I'll say is the role that three plays here the B Matrix kind of conventionally this is a transition operator that's describing how things change from time step to time time step whereas the three operator in the continuous setting is describing as we go to the right the derivative and as we go to the left the integral so like you said you have to walk back through the premises um step by step from 10 times steps forward you can go 10 9 8 let's just say but you can't skip from 10 to eight at the very least you would need a different transition operator to apply backwards similarly um you might be able to go from X double Prime take the integral and then take the integral again but you couldn't take a double integral in one jump that wouldn't be the same as these integrals okay but it's kind of like yeah there's other ways to take it like the computational irreducibility angle which is like there might be a shortcut to just apply the double integral in one step but then there's also probably equations that you like can't apply a double integral and in a way by doing a double integral in one step you've just made a single inte you know or by by going two time steps backwards here you've just taken one of a different kind of Step okay yeah because I I ran into something recently that resolved into a five integral and that's where I got confused as to how that was even possible and I'm still trying to to work through the math a little bit to understand but uh that's kind of out of the scope so I don't want to take too much time out of that if you're looking at something that's that's a really long sequence you might be able to have statistical power to talk about like the fifth derivative but you could simulate trajectories and then just plot the first derivative plot the second one plot and so on and then you'll just find that maybe there's interesting dynamics of higher orders or maybe not and then the empirical way that you could ask like how important are they would be okay so we make a real simulation that five um five actual derivatives of influence and then we make a subset models that have all five that should be able to recover 100% of the variance and then you test the one with four three two one and there's situations where even if there's really five the first derivative covers 99% of it or there might be ones where the higher derivatives do matter okay thanks here just just again just to Connected here's just evaluating the point so to say you know this is this is like the true sea levels for the next 24 hours or this the real temperature in the room so it's like if you can only have one number the best thing you can do is just say what is the temperature right now and that's this kind of first approximation and then if you were to have a second number you could say okay what's the temperature in the room and then what's the the slope and then if you had another number temperature right now and the slope and how fast it's changing and so that process is the Taylor series expansion so that's the kind of way that they talk about it in approximating functions because you can keep on adding more and more approximating terms but it can get unwieldy and you still get and yeah there's other things to say but this but I guess I'll just say this dash line does go off to Infinity like you could evaluate that dash line at this point but it' be like way way way down and then the one of the issues with using the um Taylor series is you don't know how good or how bad the dash line is except by then doing the laborous um work of checking against the actual function so like here it's like it actually is doing really well but it's not telling you it's doing well you would just find out by checking but then over here it's going to be doing absurdly bad but it's not like it's going to throw up a yellow flag and tell you I guess my question in terms of uh that that dotted line trending to Infinity is is that assumptive or is that you know provable in the sense that we just can't see past a certain State because it's you know a hidden state or is it calculable past that hidden state that is going to be infinite the dash line is very well behaved because it's a polom okay it is this polom here's the first order polinomial just a constant here's the second order it's a linear yals MX plus b basically MX plus b intercept now we're in the quadratic polom setting so it has all the kind of simple behavior of polom which is what makes it a well behaved approximator for arbitrarily unruly functions but it doesn't tell you how good it is okay also just note like on the pragmatic side pmdp the python active inference tool kit is only in discreete time at this point right I'm trying to translate some of it over too because I'm starting to work a little bit with kiss kit or I'm trying to uh to understand to try and you know use some of the functions there with Quantum Gates but that's a little bit beyond um to try and utilize that as a tool so I'm trying to put it together a little bit and then I just got a little bit confused over the last few days putting some things together so um I'll keep looking at it appreciate oh all good okay anyone else just unmute and go for it or feel free to type a question or we can look at the the upvoted questions here or any question that was written in the um chapter six or cohort six questions check okay n specific I think it's worth to kind of pull back a step and look at figure 4.2 figure 4.2 is really helpful because it kind of builds intuition for what these base graphs are in terms of like here's a situation on top left with one hidden State and one observable like there's a real temperature in the room and then there's a thermometer here's the situation with one hidden State and two observables here's a situation with two contributing factors and one observ like two things that contribute to a disease prevalence and here's a chain of causation so these are like these are the kinds of things people say like there was traffic because it was a sports game and a Friday night it's like okay that's this situation or it was a Friday night so there was a sports game so there was traffic but those are are the kinds of things that people talk about when they use language to describe causal sequences in the world and then these variables can be like anything like the observable could be any modality of observation and then the hidden States can be any latent causal Factor including things that don't like actually exist in the world like without arguing too much about whether temperature actually exists in the world you can have latent causes that are just totally proposed for better and for worse but this maps onto like a huge fraction of the kind of statistical inferences that are done and this also is kind of like if if we had like a mind map and we were just like drawing out causal factors for what we cared about like this is kind of intuitively how you would do it like if one thing was causing multiple outcomes you draw it like this if there's multiple things contributing to cause an observation you could draw it like that so this is kind of like where we touch with more intuitive Notions of cause and then that is going to get deployed in this setting where we're talking about the causal relationships between observations hidden States and actions those are the big players here and that's being unfolded through time but those are the things that we want to connect in our unified model of perception cognition and action is to a first approximation perception cognition about hidden States and action that's why those are like the major nodes in the white circles has someone tried to implement this for like motor control yes um motor control has been approached from the human motor control perspective like handwriting and movement of the arm and things like that and also the robotic motor control setting at the kind of gross level like where to move within the warehouse and at the finer scale like robotic arms and things like that Dan wasn't that related to what the Hitachi group covered I just thought that might be a good video to point to yeah good call thanks that was um guestroom 75 so here was a robotics in silico so it was not physical robots it was a simulation of robots but but this is um a group that's working with physical robots and then they um the visualizations that they show in the in the Stream were were pretty interesting and provocative I think in the sense like it it [Music] um it shows how multi-agents and also how social norms come into play so here we have basically the the red agent is the active inference agent and then it's going to be like coming on the opposite direction of the sidewalk here it's an altruistic other so the green one has the goal of making the red one go as little out of the way as possible whereas the selfish green one is trying to minimize its path so those are kind of ways in which people Implement like even these kind of social characteristics into to the preference distributions and the action Tendencies of different agents and that's kind of the journey that chapter six takes us on like which so that's kind of why the book can be read in many orders because as it's written linearly the first five chapters are like a ton of information that's the epistemic component and then the second half of the book starting with chapter six is like well here's the recipe for for building the model like what are we modeling why are we doing it what form should we use figure 4.3 so just the way the textbook is written it kind of shows us a lot of the tools in the workshop and then we get the recipe in chapter six but another way to do it and of course if you go through many times then it doesn't really matter as much where you began but another way to to go about it is start with the system of Interest Like A system that you're familiar with or curious about and then start to reverse engineer like what are the observations here like who's observing what I mean I could explore even in this Starbucks like who's observing what what are the actions that different agents are taking what preference distributions do they have you know some people want this level of sweetness or this um kind of coffee so that's like the kind of low road approach is starting more with an actual system and then building up to what formalisms come into play rather than like learning all the math equations and then just hoping that something like crystallizes out of nothing thank you Cheryl I'm gonna I'm gonna bring this in let's look at it awesome yeah the the these are great points um like well you know the the brain is a system of systems so like if we up here this could be like this is like our let's just say this is our portfolio models I mean the brain entertains multiple causal hypotheses about events so it is a system of systems and these are these are yeah good ahead I guess I meant it's not a system of just internal systems and that's where I get um mixed up I mean it's the whole of it the whole experience in the world we are not just brains we're in ecosystems we're in Social systems we're in you know what I'm saying um yeah it's like it's such a great and essential Point that's why we focus so much on the action perception Loop which initially seems kind of esoteric but then we start to see that like all agents around us are involved in it and then the only way to account for that kind of like Dynamic co-creation with the environment and the agent is to think about the interface every time step like if you're on a run or if you're driving you can't just focus on well just ignore the perception for the next time steps or just ignore the actions for the next time steps like action and perception are always happening the agent can't opt out of those happening and so it is a joint entanglement with the environment and that's like that's exactly what has to be kind of modeled to have a holistic model of even just a rat in a teamas correct correct so you're right I like it I like the systems in systems within systems model but we can only work with what we can work with that a time that does infer a certain amount of reductionism though as if if I understand it at this level I will understand it at the more at the larger level so that's a that's a an assumption that we bring to this yeah I again you know there there's no single order for going through active inference but figure 9.1 kind of captures that like here in the center is the top of figure 4.3 so this is the rat in the teamas that's the subject of our study here's the scientists in the lab so what we choose to present as stimuli are observations for the rat but they were choices by us and then the choices of the Rat become observations for us and so you could close your analysis here and say well it's just a rat in a laboratory or you could model the building around the laboratory and you could ask like how does that context matter and then you could ask well how confident are we that the the rat behavior in the lab is like it is in the field there's laboratory setups that it probably transfers and there's ones where it doesn't but just making the model of the rap behavior in the lab does not guarantee transfer and and that that's why the system of systems is so important right okay I love that but I'm not in chapter nine metab basian inference oh my gosh I don't even know I'm gonna get out of chapter four okay so I'll wait I'll wait there but um I do I do have a question have you ever heard of Marilyn Monk who's also at UCL uh she's a molecular biologist she has a really interesting um theory on uh cause Consciousness yeah and and um she takes the systems all the way from Adam to Cosmos that's what that's what got me uh thinking about what we're doing here in this in this book but uh a very interesting model that's cool like definitely some people are going that kind of like multiscale like down to as small and fast as you can imagine like Quantum and then up on through the cosmological and then this seeing this part here that reminded me of of a really great book um evolution in four dimensions which is like there are genetic epigenetic and metabolic and cultural and symbolic inheritances and there situations where one explains all of the variation but there's situations where it explains none so it's not like you can just have this in principle battle and say oh you know epigenetics wins or symbolic variation wins it's like the the field has been sketched and then the empirical question is what matters where and how and then what but those are questions to address four specific systems in practice they're not questions that can be addressed you know dusted off once and for all in principle yeah yeah yeah yeah yeah yeah I mean all right yeah there's hope for me there's hope for me then there always [Laughter] is it's I mean it's it's fun and it's a learning there's many things that that connects because like we're talking about the systems that we talk about elsewhere so when we talk about organisms we're we're we're talking about the same we're not talking about some other system and this is more like a lens that we bring to systems right a lens we bring the systems another way to think about that is like act active inference says less so we can say more but it's not like active inference is g to you know it's not a slot machine it's not going to deliver us like an account of a psychiatric or a social situation I mean the textbook is what it is like numbers are just there for people to count things let's just say so these models exist for us analyze cognitive systems but if active inference already had an answer about how cognitive systems are right that would that would curtail it from being used to explore diverse intelligences and open-endedness of creating new things yeah got it and that's why like yeah thank you that's why chapter four gives us the like the generative model it's like this is like the Linux current or like the software system but it's like but what do I do with a computer it's like well chapter 5 here's people using it to study Nels but in chapter four we see it in kind of its pure form in the general or in the abstract form which is like both sometimes very frustrating or it feels like it's like a mega infinite Learning Journey but all of the connection from ACM to any system we make that connection that connection is not for us to find Dan could you uh go back really quick for me to uh the chapter was I scroll down I think it was five where you had gone down to uh Consciousness related to that um we introduced um the differential between animal and uh trees this quote here or plants yes sorry um so in terms of they have no brain so they're called plants but in the in the context of the tree itself isn't that wouldn't that be considered um in the same way that Cheryl was just talking about different layers of the brain where the tree structure itself right has its own branches and nodes that are still connected in the same field that may grow around but again you're also deal dealing with different Dynamics where those systems will still interact given how close they are for Fitness of reaching the sun you know variably if another Leaf is over another leaf for example I totally agree like it's kind of a um it's the kind of quote that a mammal said like I think about this paper this is my PhD adviser and a colleague a framework for plant behavior and I mean now this is more of a common perspective but um there is some question whether the words to describe animal behavior such as the word Behavior or foraging can be applied to the activities of plant I mean is a root tip not foraging for nutrients in a niche oh it's slow okay but not to itself it's slow to a person running around but um so oh clant don't have Behavior they're just there but they do have behavior and then active inference helps us say yeah here's the affordances they have and the time skills over which they have them but you couldn't say like something you know doesn't have Behavior just because it doesn't have a kind of behavior right well that go ahead go ahead sorry go ahead no go ahead Char sorry I I was just saying that um Behavior Uh assumes Consciousness and Oxford dictionary defines it as the state of being aware and responsive to one's surroundings so if you like Marilyn monk she talks about awareness is sensing so if we say sensing and responsive to one surrounding then we have to say plants have a Consciousness according to Oxford dictionary so what you're what you're saying saying about the leaf and the tree and the root tip um brings the word Consciousness here which is problematic for me because it has no end I mean what doesn't sense respond so I guess say even rocks do that that's that's these are the mega philosophical questions like does pan cognitivism the idea like we approach all things as cognitive systems or that they truly are cognitive systems does pan cognitivism imply pan psychism is there something it's like to be anything because all things are engaged across the interface in sensing and adapting even something that's inert it's still is bounded by an interface and there's still something on the other side of the interface even if it's just a cup of water but then is there something it's like to be a cup of water because it'd be different if the cup of water weren't there so it's doing something in relationship to its Niche and so that's people looking for continuums where it looks like there's sharp distinctions and looking for Sharp distinctions amidst the continua like it has to have a higher order awareness like it has to be aware that it's aware like our attention is not only about Target of our attention we're also aware that we're paying attention to something and this is where so many of the people bring in like meditation and the mindfulness connections and then some people use that to differentiate or distinguish human cognition from other types other people deploy like literally the same formalisms to argue for the continuity of human cognition and then this has been oh yeah go ahead um has there been discussion around like how much our sensors um contribute to our cognitive function say I mean just just as a mind experiment you if you took a human brain and you put into a dog's body would we start thinking like a dog more like people have thought wild stuff well I mean just to come at that from a different perspective I was thinking about this the other day in terms of cybernetics right like if we get to a point where we're able to do a full transplant of you know a mind for example into a cybernetic system wouldn't that be the same argument you know would would you have the same kind of organ rejection for example um but it would would it be and I guess that would be very assumptive at this point but like would the the false body reject the mind or would the Mind reject the false body in that sense I was thinking of more from this perspective of you know how necessary is it for intelligence or like general intelligence like you know do want if want to get high level's intelligence do we need a an embodiment system that you know senses the world um in in a complex way that you we have you I think we have like uh 100,000 like Sten receptors or something our body and it's continuously feeding into our body for many different layers and you know maybe that's something that that's crucial to you know the feelings and emotions that we have um you because there theories where you know like our our stomach itself contributes to our you know cognitive function so you know I wonder there's been discussion there from you kist and or in the active influence Community because the um you know this community seems to think more about um I think the holistic um yeah just a bigger picture in terms intelligence so I did my PhD at the University of Virginia in the United States and there was a little known secret there of a guy who studied near-death experiences reincarnation from all over the world Ian Stevenson was his name uh they didn't even really talk about him until years and years decades later um he was a well-known secret but his point was that uh Consciousness can exist without a body and even without a brain and at that time it really got me yeah there he is uh it really got me to thinking about this whole idea of Truth um there is a scientific model of Truth in which of course this is done by humans but before there were scientists there were still people seeking truth and even in Act of inference they're called the mystics um and then I think in our day you know there there are poets who kind of intuitively or think deeply about something but it's still a search for truth so I think um for me it has been useful to acknowledge these different ways and perspectives and modalities and not think one is greater than the other that's a great Point there's so much to say it it reminds me of like there's the parable of the the blind people and the elephants and they're like touching different parts of the elephants okay now we could kind of modify that into one person can see one can hear one can smell one can talk so there's different modalities that have not just touching a different part of the body but they're actually getting like a different wavelength they're getting totally different modalities of the elephants um that's that's one interesting component and then one other point is like um so earlier asked like what is the role of the sensation and perception in intelligence there's a ton of answers but one answer that like Adam saffron and others have pursued is um the kinds of relationships that our sensory apparatus require to solve help us build architectures that reflect the structure of our reality like grid cells in the brain those evolved because we were moving in spaces with certain geometries but then those became later utilized for other kinds of arbitrary cognitive navigation or like having hierarchically nested perceptual apparatus it arose to do something that it was repurposed to do later because that Motif is so effective at the kind of statistical challenge that it evolved to solve or address and there's probably no end to those motifs wow yeah it's super it's like chapter four is so hard this is where it has to take us can I I know over time can can I ask like a very fine Grand question to zoom all the way back in yeah yeah um so I I kept getting confused in the text by some of the conventions in the math um are bold faces and our serifs versus S serifs meant to mean anything so if you look at equation 4.11 for example we have caps SF f equals bold pi times bold SF F and I I I looked around I couldn't figure out like what if anything are those like changes in the specifics of how things are rendered meant to indicate I'm totally with you that's one reason why like having the um having the written equation form I feel like is really vital because even even experts will not be able to converge upon this kind of a description without looking at the same page there's just too many ways to read even the equations and that's even before you get into like bold and italics and like font and stuff like um it looks like bold is only mentioned one time in the supplement or appendix B referring to it as being like a sufficient statistic but I don't think that's how it's being used above I feel like bold is being used to describe whether it's a vector or not but I I I don't know and like faced with that daunting of trying to sift through where they or if they Define it for like typographical phenomena that I don't even necessarily know the name of that's why I'm very excited for ontology driven natural language descriptions of the equations and then we can render them to look however we want so we could render them with a full English word or we could render it with a full non-english word or we could use a python variable because I I don't even if you could dig to the bottom and find out if they defined how they use notation other papers are going to do different notation so it'd be kind of like a a local victory yeah at this point I would be satis or I would be helped by local Victory there's like a couple of pages where they like seem to use this distinction a lot and I was like not following their choices at all um but but yeah I appreciate that like a shared resource like helps create some clarity um but uh yeah okay equation 411 uh yeah is an example and then like above and below this there's a few there's like a number of others that are uh I think in to use the same distinction and I I have to assume that the authors had some purpose in mind so I don't think it's just like Matrix versus Vector versus scalar and the question is what does the Bold typing versus the italics mean what does what does the bold mean and what does the serif versus un serif mean okay and if if no one knows I guess no one knows and that that's the state of the world but yeah I was I was what do you mean by S you mean this F versus this F right so they are distinguished by by the Bold feature but like other things are are also distinguished by the Bold feature but not by the SF versus SF change okay it seems like we always have a co-occurrence of bold and straight and italics is never found as bold so I'm tempted to say that bold and straight means sufficient statistics of whereas the italics is referring to the underlying variable as it is but we can like literally just write the question down in one of the rows and then whatever we can't resolve we can email friston at all okay fair enough yeah yeah yeah I mean I dream of the day when we can Mouse over and have multiple modalities of what these are and and like have someone says I want to Mouse over and see examples someone else says I want to Mouse over and see like the programming variable I I don't know if there's another way to make like a fixed artifact that that could speak to so many people in so many different situations okay all right well interesting times question uh yeah yeah um I then coding exercises that I can go for to um understand these um equations better yes i' go go over the code page here's several dozen um code uh examples like when you see and then also I would highly recommend the PDP documentation because like when it's shown as the base graph and it's like it seems like again it seems like it's going to be like this math and analytical thing but when you go to active inference from scratch you you define ABCD and then you have all of the steps all of the utility functions are are defined and then um you have the active inference Loop and then once you've gotten to here here's active inference Loop A B C D and the environment and the time Horizon so if you've um run up at the equations Andor you want to try a little bit more of like a bottomup computer science or computer engineering approach this can be really clarifying but those are very useful with PDP um but any of the code examples will be fun or useful however pmdp and RX and fur H um have the best documentation so and we're here to make the the material more pedagogical so if there's like a example that you're excited about like Lally let's have the authors do a live stream or let's make the code better annotated or documented or whatever we need to do um do they have any examples where they're doing um active influence on just some basic um basic predictions like I don't know tempature or yes like um the uh RX infer documentation has has like a range of examples um ranging from like don't worry about something being too simple truly like even the examples um of coin flipping are a great place to start but this is a ton of information about all kinds of examples and then usually the people who are making the packages are focused on making the package if not like working three jobs so if something's missing I mean change the world or change your mind so if you think there could be a cool example that's what we're doing here so I hope that we can make it if it's not already there yeah wow that was chapter 4 the next two weeks we'll come back for chapter five and look at more examples in the Maman nervous system and then we'll have just a final meeting kind of catch up hopefully get people's feedback and connect on projects and just see where things are at in The Institute the ecosystem so thank you all see you next time thank you bye bye thanks everyone thank you